

POLICY ANALYSIS · CCUS SERIES, PART II · APRIL 2026

India's CCUS Gap Is Not About Money

The Union Budget allocated Rs 20,000 crore. A national roadmap was published. The obligated industry base is real and growing. And still, the sector cannot move. The obstacle is not capital. It is the absence of five regulatory instruments that no allocation can substitute for.

THE CLIMATE DESK · BANGALORE · APRIL 2026

Start with the question that the capital cannot answer. Before India can scale carbon capture, does it actually have a commercially viable catalyst or solvent to capture with?

Everyone is talking about upscaling and financing. The critical question we are ignoring is: do we really have a catalyst or solvent which is commercially viable? My answer is no.

PROF. VIVEK POLSHETTIWAR · TIFR; MEMBER, DST CCUS HIGH-LEVEL TASK FORCE

That observation from one of India's leading nanocatalysis researchers is the frame for everything that follows. The technology gap is real. Amine-

based capture remains commercially problematic after decades of use. Membranes have unresolved permeability issues. The Rs 20,000 crore CCUS mission announced in Union Budget 2026-27 cannot fix any of that. What money can fix — and has not — is the regulatory architecture that determines whether anyone builds anything at all.

Rs 20,000 Cr CCUS MISSION, UNION BUDGET 2026-27	<Rs 2,000 Cr ESTIMATED ACTUAL DEPLOYMENT BY MARCH 2027	750 MTPA DST CAPTURE TARGET BY 2050
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There is no specific plan for how the Rs 20,000 crore will be utilized. I expect less than Rs 2,000 crore actually deployed by March 2027.

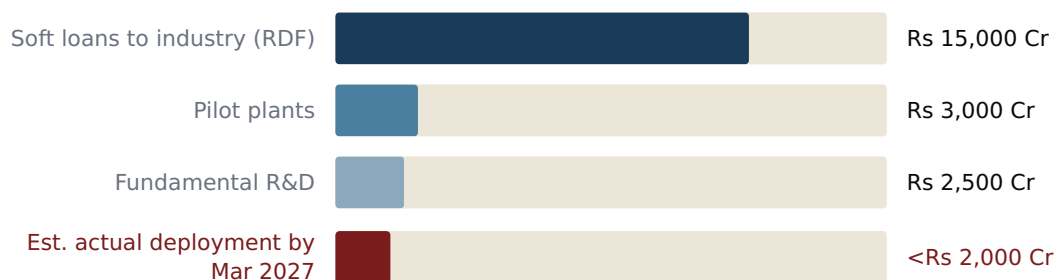
V. SUBRAMANIAN · FORMER SECRETARY, MNRE; BOARD CHAIR, GPS RENEWABLES

The allocation is structured as three buckets: approximately Rs 2,500 crore for fundamental research, Rs 3,000 crore for pilot plants, and Rs 15,000 crore in soft loans to industry via the Research Development Fund. These are categories, not milestones. Industry cannot plan capital expenditure against categories. Five specific regulatory instruments are missing. Here is what they are.

A Deployment Plan, Not Just an Allocation

MISSION ALLOCATION BREAKDOWN

Where the Rs 20,000 Crore Is Supposed to Go — and What Will Actually Move



Source: DST CCUS R&D Roadmap, December 2025. V. Subramanian, Former Secretary MNRE.

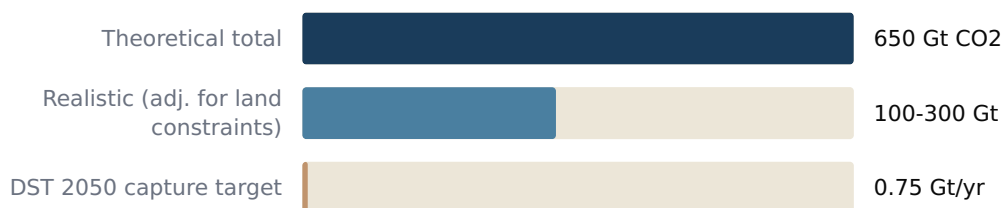
The DST roadmap is a phased plan. The mission allocation is not. What is needed is a published milestone schedule with annual targets by sector and technology, updated publicly each year. Thailand's industrial decarbonisation policy is the operative model: year-by-year tracking, specific targets, accountability built into the architecture rather than added as an afterthought. Without this, the Rs 20,000 crore risks the pattern that has defined many Indian climate allocations: announced at the headline figure, deployed at a fraction of it.

Geological Storage Has No Legal Owner

Who owns the pore space below the ground in India? The answer is: no one, legally. There is no legislative instrument that establishes subsurface pore space ownership, separates it from surface land rights, or assigns liability for CO₂ stored underground. This is not a technical gap. It is a precondition for financial close. No lender will fund a sequestration project where the asset being created has no legal title.

INDIA'S GEOLOGICAL STORAGE POTENTIAL

Theoretical Capacity vs. Realistic Estimate vs. 2050 Target



Source: DST CCUS R&D Roadmap, December 2025. Sushma Rawat, Former Head of Exploration, ONGC.

We are trying to save one part of the environment while potentially creating another problem. The risk of CO₂ migrating and contaminating potable water sources requires extensive geological modelling before any injection begins.

SUSHMA RAWAT • FORMER HEAD OF EXPLORATION, ONGC

The geological potential is not in question. What is missing is the legislation. Subsurface pore space law exists in the US, Australia, and

across the EU. Parliament needs to resolve three specific questions: who owns the pore space, who bears liability for CO2 leakage over decades, and what monitoring, reporting, and verification framework governs stored CO2 certification. None of these require novel invention. The precedents exist.

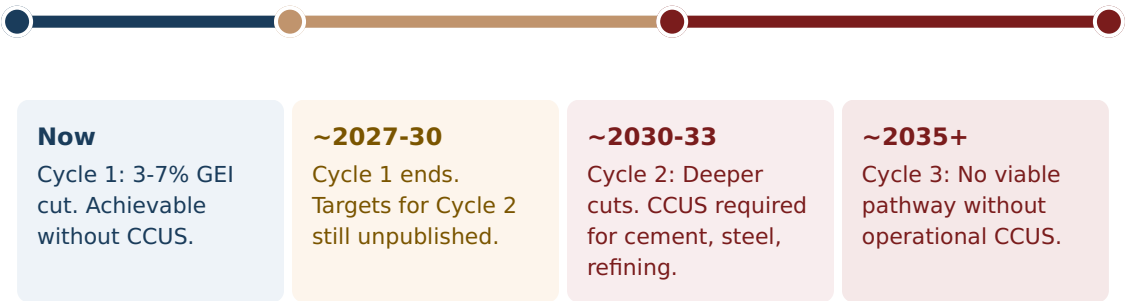
GAP 03

Industry Needs to Know What Cycle Two Looks Like

India's Greenhouse Gas Emissions Intensity targets under the Carbon Credit Trading Scheme are currently binding for approximately 500 industrial entities. First-cycle reductions of 3 to 7% against baseline are manageable without CCUS. Companies can meet them through efficiency improvements and fuel switching. This is creating a dangerous illusion of adequacy.

CCTS COMPLIANCE TIMELINE

When CCUS Transitions from Optional to Unavoidable



Source: Dr. Ashish Lele, Director CSIR-NCL; Chair, DST CCUS High-Level Task Force. MoEFCC CCTS framework.

The real challenges begin in the second and third cycles. Industry has a 5 to 7 year window to build CCUS capability. If planning does not start now, they will not be ready in time.

DR. ASHISH LELE · DIRECTOR, CSIR-NCL; CHAIR, DST CCUS HIGH-LEVEL TASK FORCE

Industrial capex planning on CCUS runs on 7 to 10 year horizons. A cement plant that must be CCUS-capable by 2033 needs to begin engineering design now. BEE and MoEFCC need to publish indicative second and third cycle GEI trajectories immediately. Indicative is sufficient for planning. Absent is not.

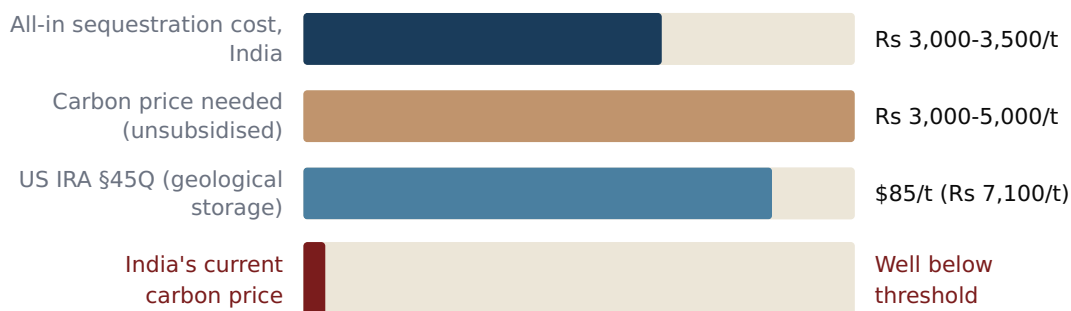
GAP 04

The Economics Need a Bridging Instrument

The all-in cost of sequestering one tonne of CO₂ in India, including capture, liquefaction, transport, and injection, lands at approximately Rs 3,000 to Rs 3,500. The carbon price required for a private investor to finance that chain without government support is Rs 3,000 to Rs 5,000 per tonne, roughly \$50. India's current effective carbon price is well below that level. The gap between cost and available price signal cannot be closed by enthusiasm for the technology.

THE SEQUESTRATION ECONOMICS GAP

Cost of Sequestration vs. Carbon Price Required vs. Global Incentive Benchmarks



Source: Prof. Rajnish Kumar, IIT Madras. Vishnu Rajeev, Speciale Invest. IEA Policy Database: US IRA \$45Q, 2022. IEEFA European CCS cost estimates, 2025.

GLOBAL POLICY INSTRUMENT COMPARISON

What Major Markets Have Built That India Has Not

COUNTRY	INCENTIVE INSTRUMENT	PER-TONNE VALUE	PORE SPACE LAW	MRV FRAMEWORK
United States	IRA \$45Q tax credit	\$85/t geological	● Yes	● EPA Subpart RR
Norway	Longship government grant	Full value chain subsidy	● Yes	● Norwegian Offshore Directorate
European Union	EU ETS carbon price	~60-80 EUR/t	● CCS Directive 2009	● CCS Directive MRV
Australia	Safeguard Mechanism	A\$2.55bn allocated	● Offshore Petroleum Act	● CER framework
India	Rs 20,000 Cr mission (loans)		● None	● None

COUNTRY	INCENTIVE INSTRUMENT	PER- TONNE VALUE	PORE SPACE LAW	MRV FRAMEWORK
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No per-
tonne
incentive

Source: IEA Policy Database. US Congressional Research Service §45Q, 2023. Equinor Northern Lights, 2025. EU CCS Directive. Australian DCCEEW. DST CCUS R&D Roadmap, December 2025.

We need to get to at least Rs 5,000, or \$50, per tonne. Without a carbon market mechanism at that level, I have very high hesitation to invest in CCUS infrastructure.

VISHNU RAJEEV · PARTNER, SPECIALE INVEST

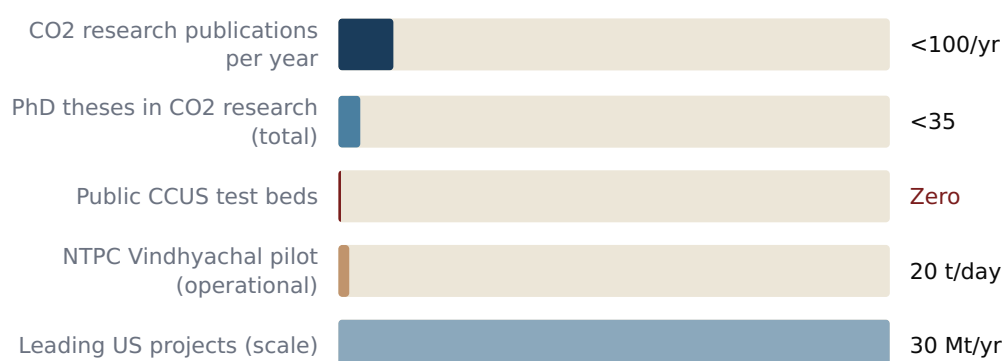
Norway's Northern Lights project, which began injecting CO2 in August 2025 with Phase 1 capacity of 1.5 million tonnes per year, was financed precisely because government subsidy bridged this gap. The US 45Q credit, raised to \$85 per tonne under the Inflation Reduction Act of 2022, did the same. India can structure an equivalent instrument as a production-linked incentive, a direct tax credit, or a viability gap mechanism. The design is not the constraint.

The Technology Must Be Tested Before It Can Be Scaled

India has no open-access semi-industrial test facility for CCUS. Companies wanting to pilot capture or conversion technology at meaningful scale must rent facilities abroad, at high cost, on emissions profiles that do not match Indian industrial conditions. The result is that every technology validated offshore is an opportunity to build Indian IP and operational expertise that did not happen here.

INDIA'S CCUS RESEARCH BASE

The Infrastructure Gap Behind the Technology Gap



Source: Dr. Ashish Lele, CSIR-NCL. NTPC Vindhyachal: Down to Earth, February 2026. US scale: Global CCS Institute, 2025. Journal of Geological Society of India, November 2025.

India built its pharmaceutical and biotech export capability on shared research infrastructure, publicly seeded and independently operated. Startups accessed facilities without bearing full capital cost. Large companies paid flat fees. That model produced one of the world's most competitive generics industries. The energy sector has no equivalent. A semi-industrial CCUS test facility, with a CO2 source, all utilities, and pre-

cleared regulatory approvals, would cost a few hundred crores to build. That is a rounding error within the Rs 3,000 crore pilot allocation. The governance model is the only unresolved question: it must be independently operated, not managed by a government institute or a large corporate.

For me, evaluating CCUS startups, the biggest trouble has been showing how the fundamental unit economics work. People are not able to demonstrate a pathway for how the CAPEX will come down to something commercially viable.

VISHNU RAJEEV · PARTNER, SPECIALE INVEST

THE FIVE GAPS

What India Must Resolve Before Capital Can Move

POLICY READINESS ASSESSMENT

Status of Five Critical Regulatory Prerequisites

01

PARTIAL

Milestone-based deployment plan for the Rs 20,000 Cr mission

Allocation structure exists. No published annual targets by sector or technology. No public tracking mechanism.

02

MISSING

Subsurface pore space legislation

No law establishing ownership, liability, or MRV for geological CO₂ storage. Equivalent exists in US, Australia, EU.

03

MISSING

Indicative second and third CCTS GEI cycle targets

First-cycle targets published. Second and third cycle targets not signalled. Industrial capex planning requires 7 to 10 year visibility.

04

MISSING

Per-tonne sequestration incentive

No bridging instrument between sequestration cost (~Rs 3,000-3,500/t) and investable carbon price. US offers \$85/t. Norway subsidises full chain. India: nil.

05

MISSING

Open-access semi-industrial test infrastructure

Zero public test facilities in India. Companies pilot abroad. No independently operated shared infrastructure within the pilot plant allocation.

Source: DST CCUS R&D Roadmap, December 2025. MoEFCC CCTS framework. Conference proceedings, March 2026. IEA Policy Database.

None of these require India to invent instruments that have no precedent. The legal frameworks, incentive structures, and shared infrastructure models all exist elsewhere. What they require is that the government moves from announcing the mission to administering it.

If the cost of capturing 1 kg of CO₂ is Rs 4, and we emit 0.8 kg of CO₂ per kg of cement, the cost of cement goes up by 80%. Is the end consumer in India ready to absorb that cost? And without a government procurement preference for low-carbon cement, there is no market incentive to find out.

LOVISH AHUJA · CHIEF SUSTAINABILITY OFFICER, DALMIA BHARAT

The second GEI compliance cycle arrives in the early 2030s. The industries that have built CCUS capability by then will hold a structural cost advantage over those that have not. Industrial capex planning on that timescale starts now, or it does not start in time. The five gaps above are not obstacles to ambition. They are prerequisites for it.

ABOUT THIS REPORT

The Climate Desk covers India's carbon markets, climate policy, and energy transition. This article draws on primary data from the DST CCUS R&D Roadmap (December 2025), Union Budget 2026-27, the IEA Policy Database, US Congressional Research Service §45Q analysis, Equinor Northern Lights project documentation, and statements by senior researchers, industry leaders, and investors in the CCUS sector.

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